

LUKAS LEIBOLD — GEOTECHNICAL RESILIENCE PROJECT · COMMUNITY PREDICTION COMPETITION · 2 MONTHS TO GO

Submit Prediction

Predict the cumulative grain size distribution of a soil sample from photographs of its surface.



Overview Data Code Models Discussion Leaderboard Rules Team Submissions

Competition Host

Lukas Leibold — Geotechnical
Resilience Project

Prizes & Awards

\$500
Does not award Points or Medals

Close

2 months to go

Participation

1,278 Entrants
496 Participants
470 Teams
5,163 Submissions

Tags

[Submit Prediction](#)

This competition is hosted as part of the EU research project Geotechnical Resilience via Intelligent Design and Boku Unierversity

Does not award Points or Medals

Close

2 months to go



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Submissions are evaluated using the logarithmically weighted Earth Mover's Distance (EMD), also known as the Wasserstein-1 distance, between the predicted and the true cumulative grain size distribution at the eleven fixed support points defined by DIN EN ISO 14688-1. For a single sample, the EMD is computed as

Predicting Soil Grain Size Distributions from Images

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$$EMD = \sum_{i=1}^{10} |F_i - \hat{F}_i| \cdot (\log_{10}(x_{i+1}) - \log_{10}(x_i))$$

where F_i is the true cumulative mass fraction (%) at support point i , $\hat{F}_i$ is the predicted value, and x_i is the grain diameter (mm) at that point. The sum runs over the ten intervals defined by the eleven support points:

0.002, 0.0063, 0.02, 0.063, 0.2, 0.63, 2.0, 6.3, 20.0, 63.0, 200.0 [mm]

The logarithmic weighting matches the conventional log-scale representation of grain size in geotechnical practice, so errors in the fine range (e.g. 0.002 mm) count as much as errors in the coarse range (e.g. 20 mm). Geometrically, the EMD is the area between the true and predicted cumulative grading curves on a log scale.

The final score is the mean EMD across all evaluated samples. Lower is better. The metric takes values in the range [0, 500], where 0 indicates a perfect prediction at every support point.

Baseline.

A trivial baseline that ignores the images is provided. It predicts an equal mass share ($\approx 9.09\%$) for each of the eleven grain-size bins, giving the same cumulative curve for every sample:

sample_id, 0.002, 0.0063, 0.02, 0.063, 0.2, 0.63, 2.0, 6.3, 20, 63, 200

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```
sample_id,0.002,0.0063,0.02,0.063,0.2,0.63,2.0,6.3,20,63,200
```

TEST_XX, 9.09, 18.18, 27.27, 36.36, 45.45, 54.55, 63.64, 72.73, 81.82, 90.91, 100.00

Public and Private Leaderboard

During the competition, only the public score is visible. The final ranking is determined by a weighted combination of both:

$$\text{Final Score} = 0.30 \cdot \text{Score_public} + 0.70 \cdot \text{Score_private} **$$

Submission File

For each sample in the test set, predict the cumulative mass fraction at the eleven support points. The file must contain a header and have the format of the provided submission csv.

sample_id	0.002	0.0063	0.02	0.063	0.2	0.63	2.0	6.3	20	63	200
	2.1	5.3	12.4	28.7	51.2	68.4	81.3	91.0	97.2	99.1	100
	0	0	3.1	9.7	22.3	63.4	87.3	93.0	99.2	100	100
ect.											

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A submission is valid if all values lie in the interval $[0, 100]$ and form a non-decreasing sequence across the eleven support points (cumulative mass fractions are monotonic by definition). The value at 200 mm should be 100.00, since none of the soil samples in the dataset contain particles larger than 200 mm. Submissions that fail these structural checks are automatically rejected.³

Prizes



The best 3 participants will receive a monetary price.

Total Prizes Available: 500\$

- **1st Place** - 300\$
- **2nd Place** - 150\$
- **3rd Place** - 50\$

Citation



Lukas Leibold, Enrico Soranzo,. Predicting Soil Grain Size Distributions from Images. <https://www.kaggle.com/competitions/soil-grain-size-from-photos>, 2026. Kaggle.

Cite

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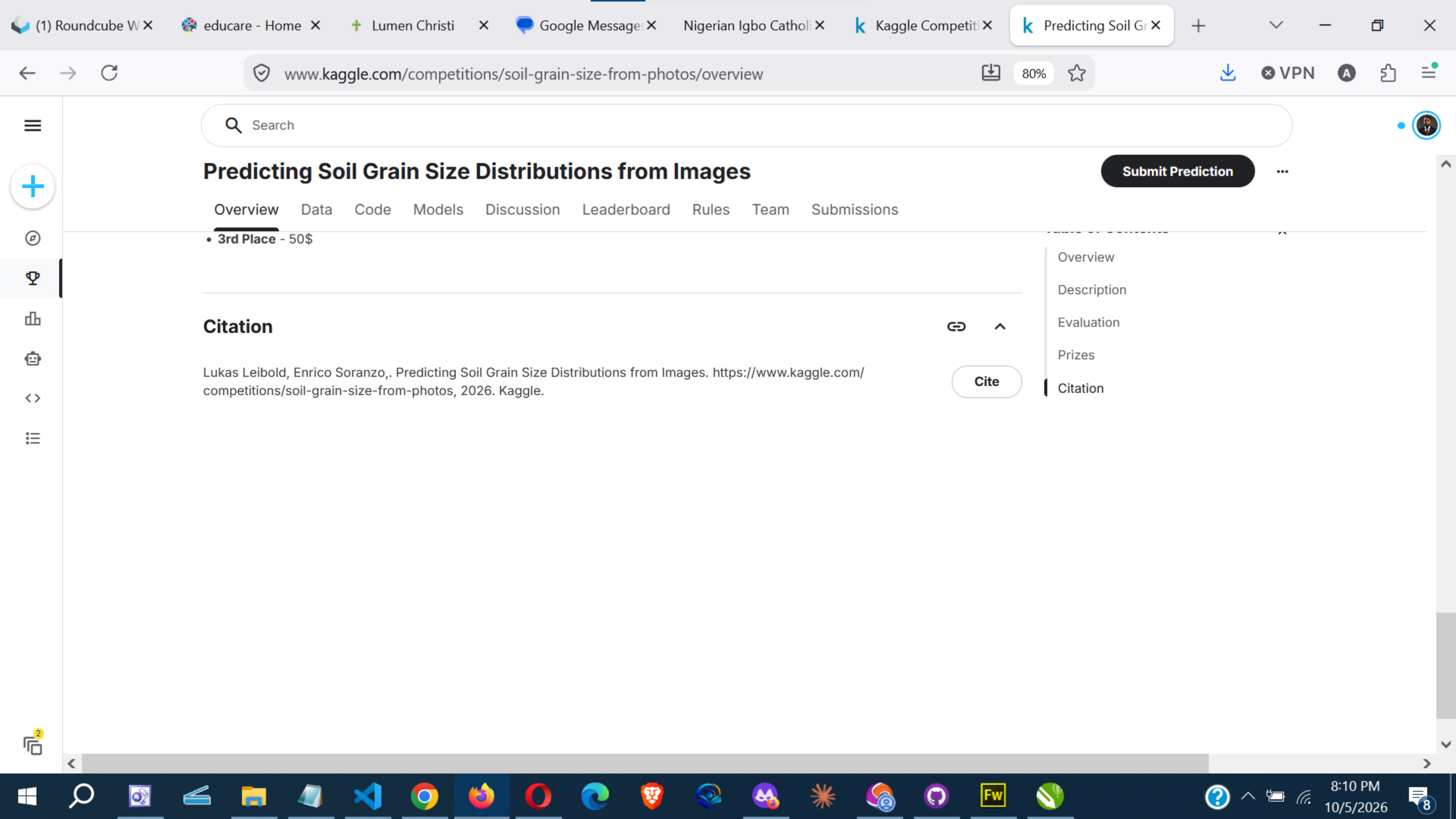
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• 3rd Place - 50\$

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